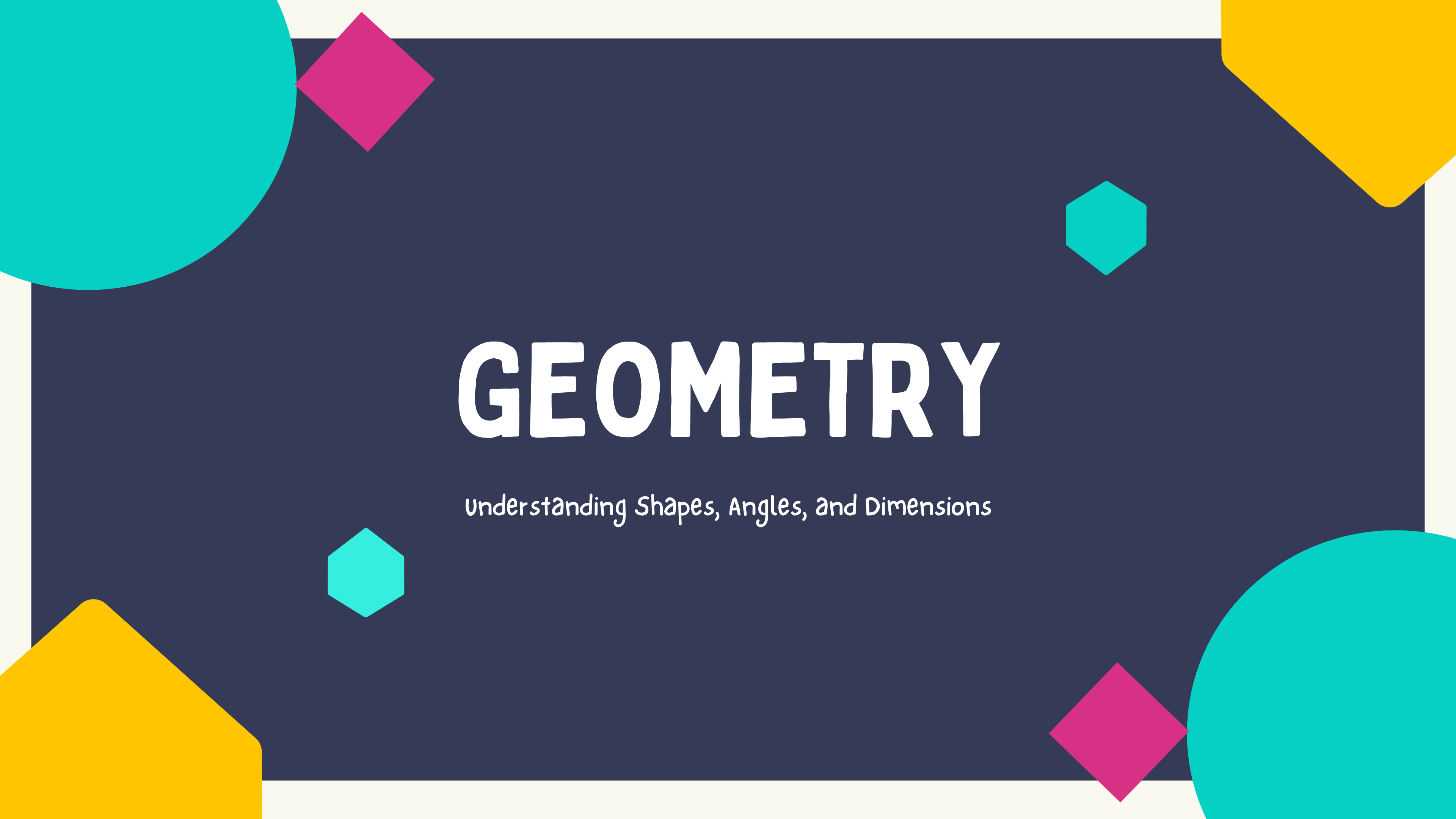




GEOMETRY

Understanding Shapes, Angles, and Dimensions

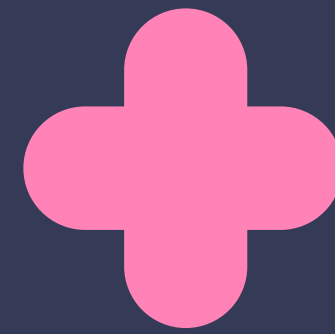
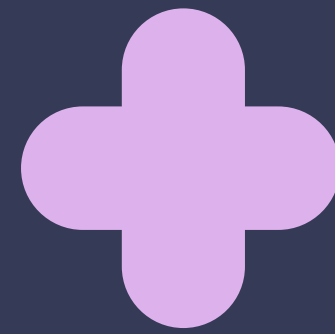
WHAT IS GEOMETRY?





DEFINITION

- Geometry is a branch of mathematics that studies shapes, sizes, angles, and dimensions of objects.
 - It helps us understand the world around us, from the structure of buildings to the patterns in nature.
- 
- 

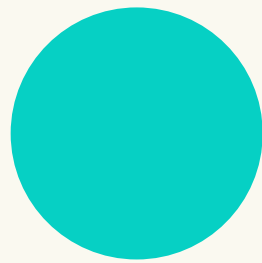


BASIC CONCEPTS IN GEOMETRY

Introduction to fundamental elements in geometry.



POINTS, LINES, AND PLANES



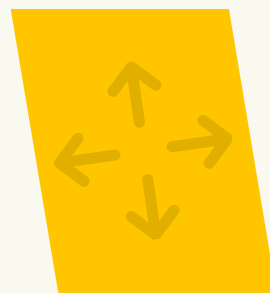
Point

A location in space with no size (denoted by a dot, like A).



Line

A straight path that extends infinitely in both directions.

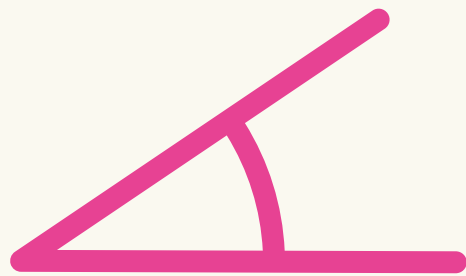


Plane

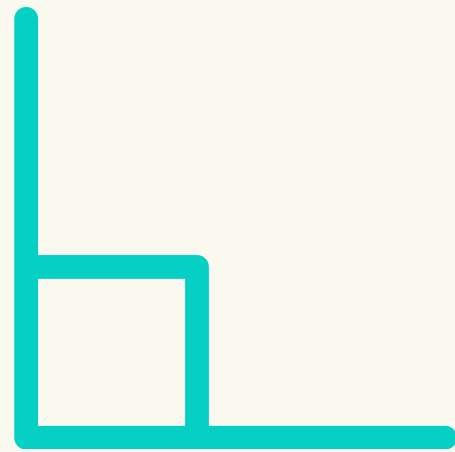
A flat surface that extends infinitely in all directions.

ANGLES

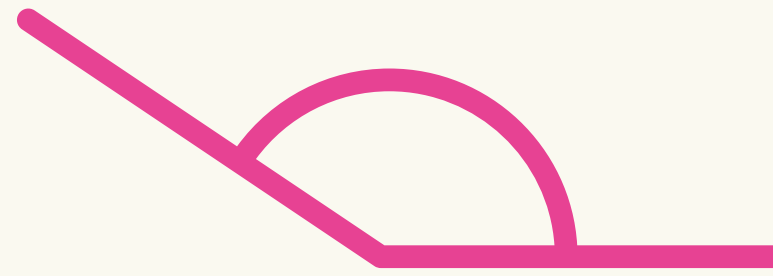
- Formed when two lines meet at a point.
- Types of angles:



Acute angle ($< 90^\circ$)



Right angle (90°)

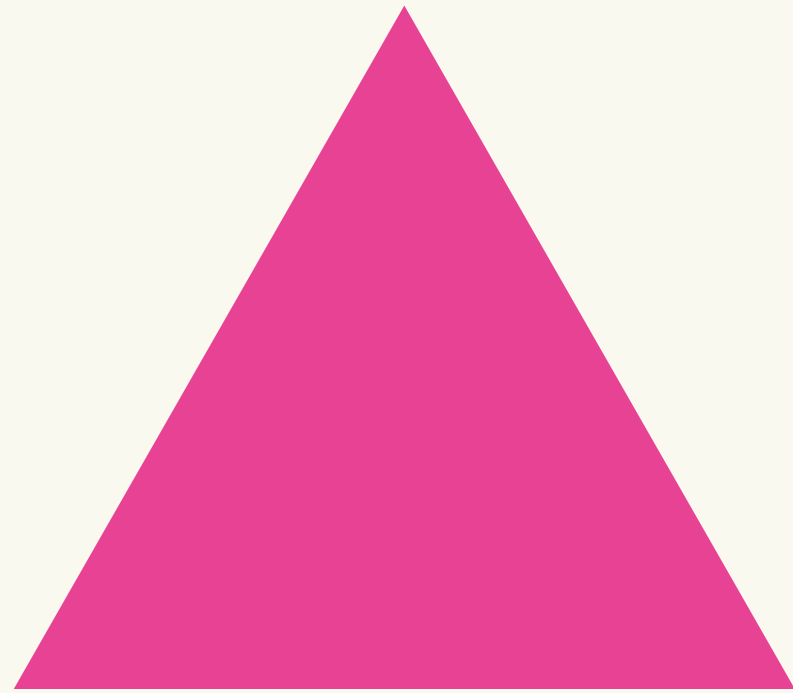


Obtuse angle ($> 90^\circ$ and $< 180^\circ$)

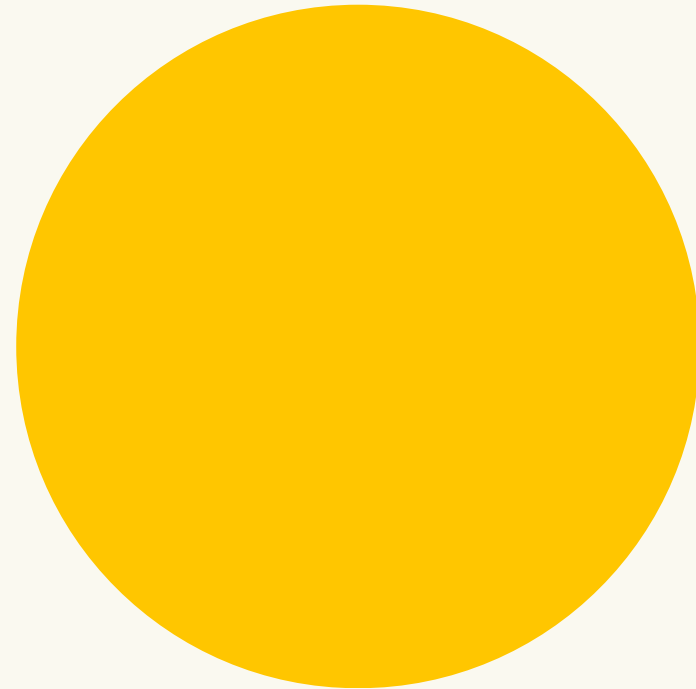


Straight angle (180°)

2D SHAPES (FLAT SHAPES)



Triangle (3 sides)

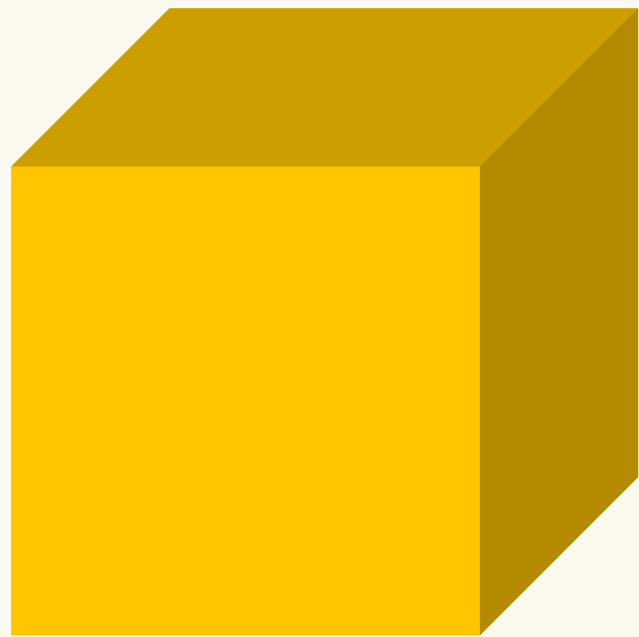


Circle (A round shape with no sides)

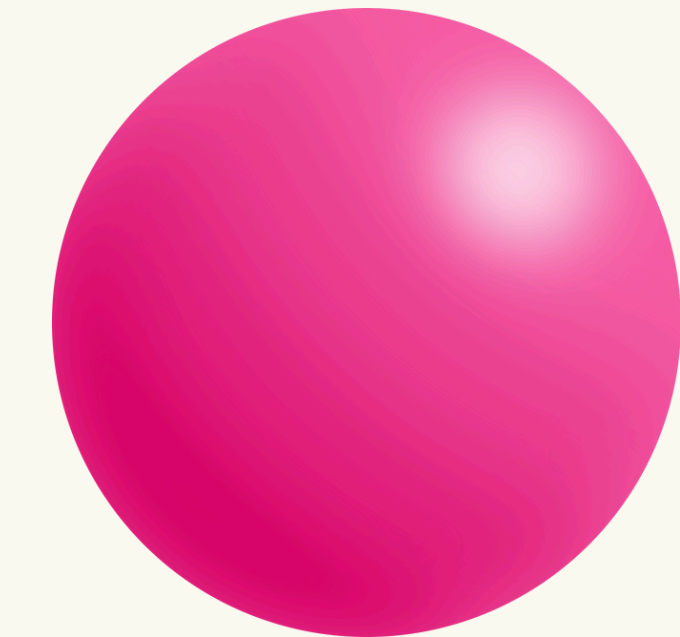
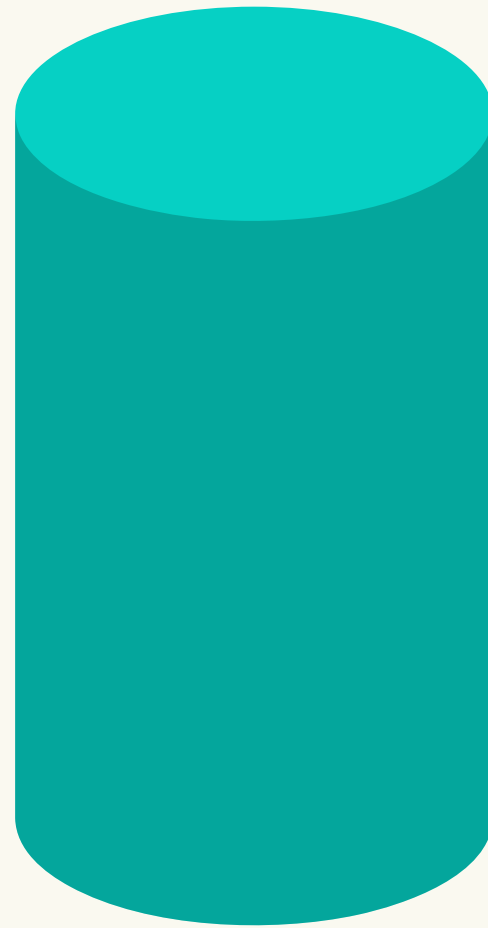


Quadrilateral (4 sides)

3D SHAPES (SOLID SHAPES)

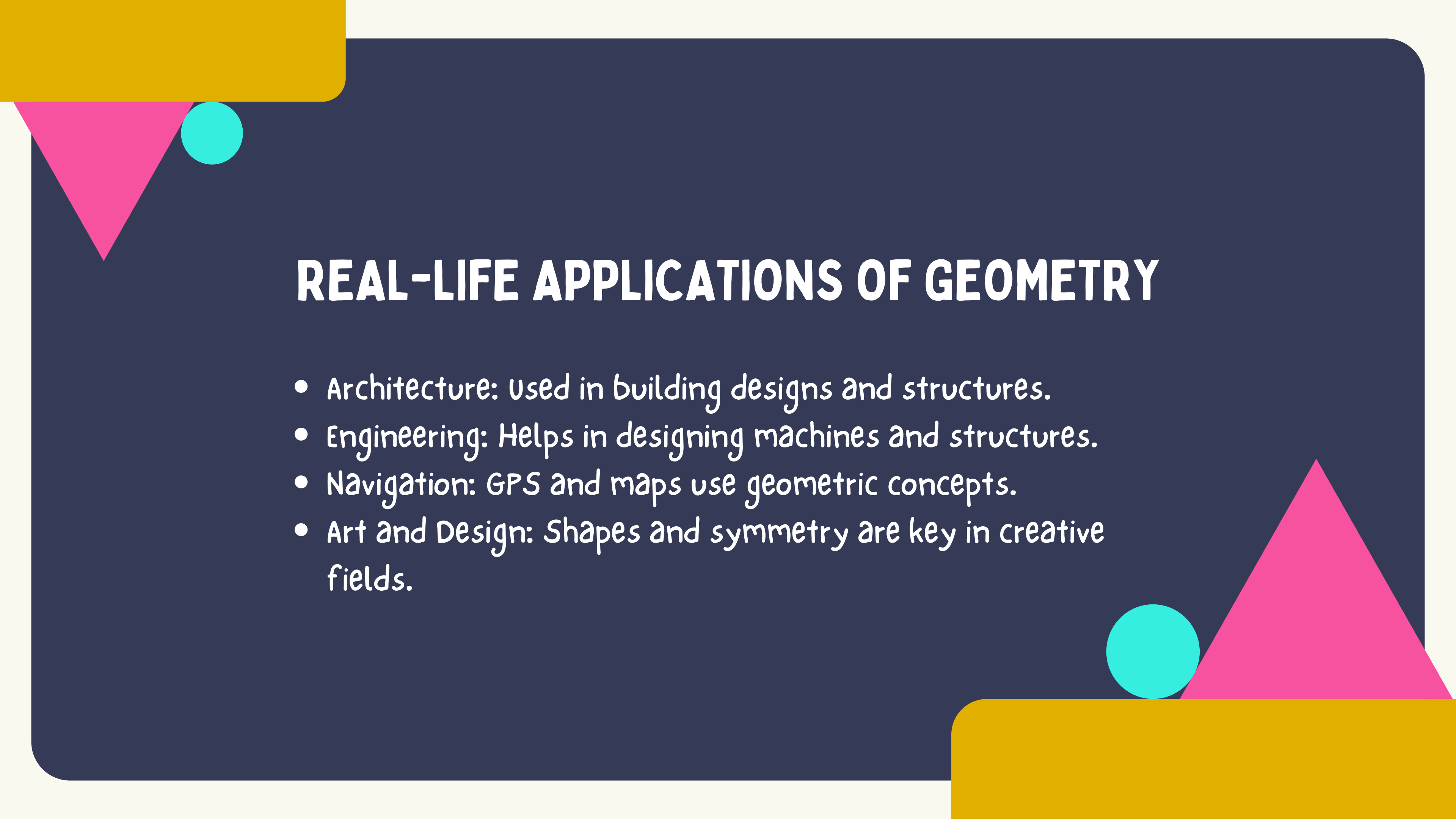


Cube (6 square faces)



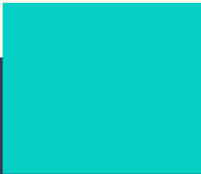
Sphere (Perfectly round)

Cylinder (Circular base
with a curved surface)

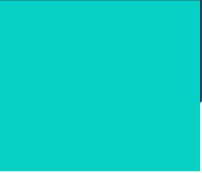


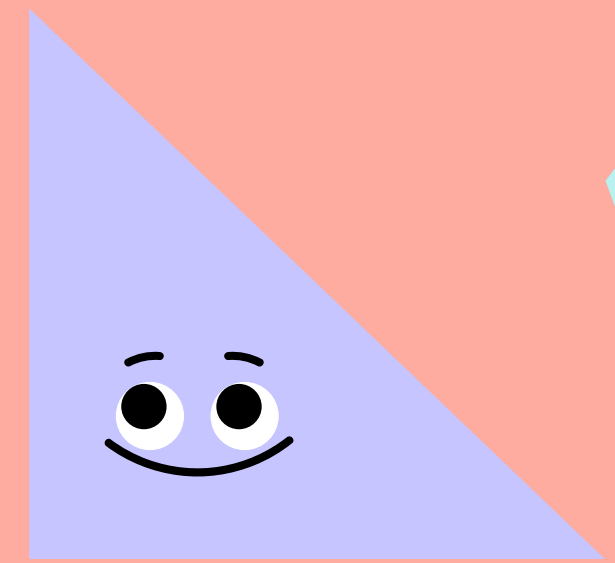
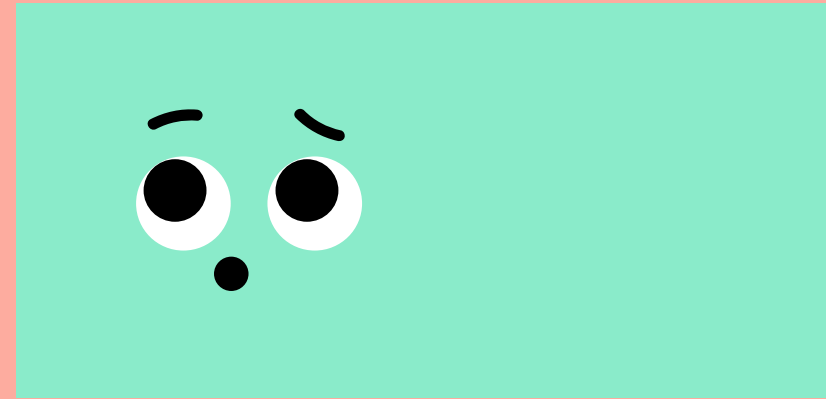
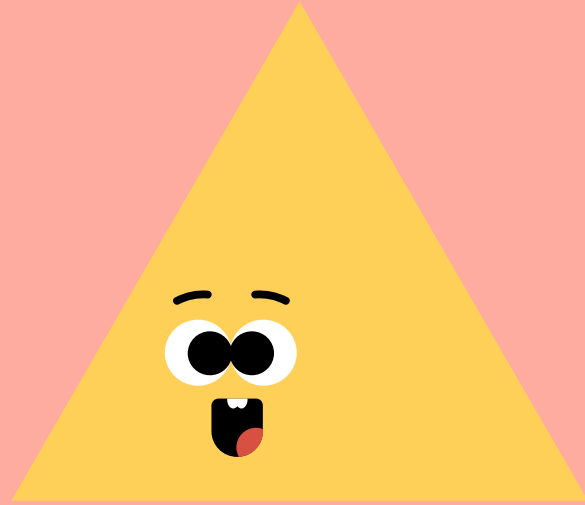
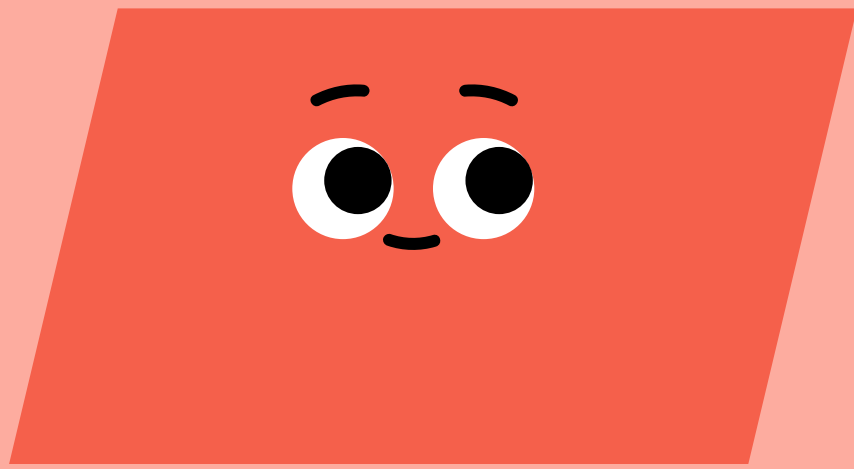
REAL-LIFE APPLICATIONS OF GEOMETRY

- Architecture: Used in building designs and structures.
- Engineering: Helps in designing machines and structures.
- Navigation: GPS and maps use geometric concepts.
- Art and Design: Shapes and symmetry are key in creative fields.



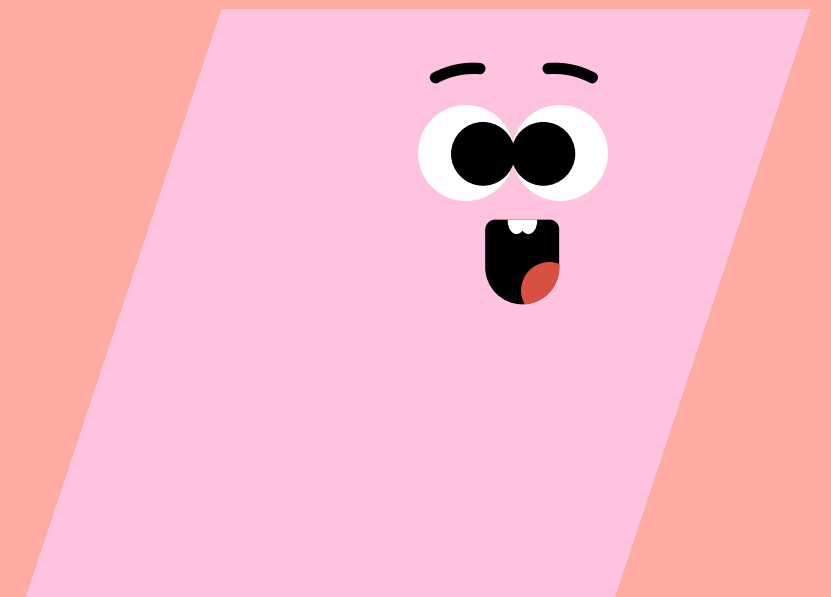
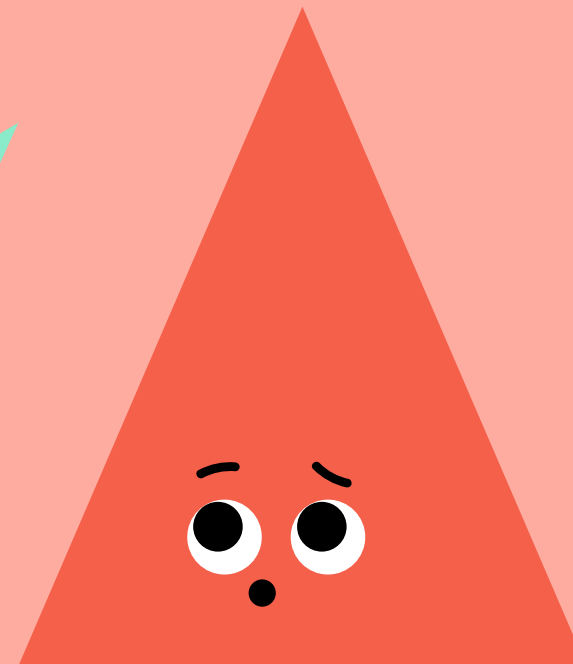
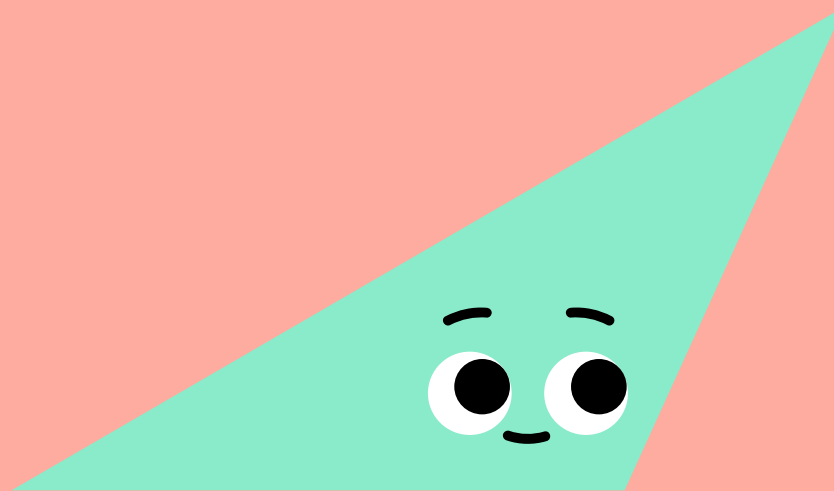
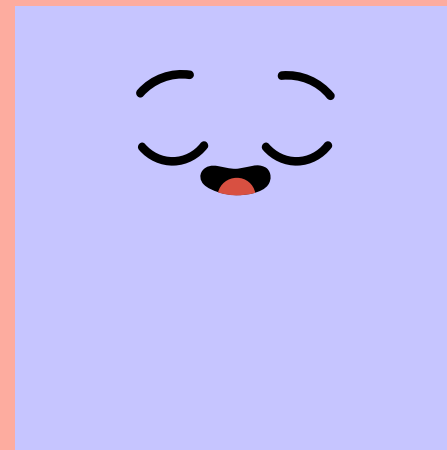
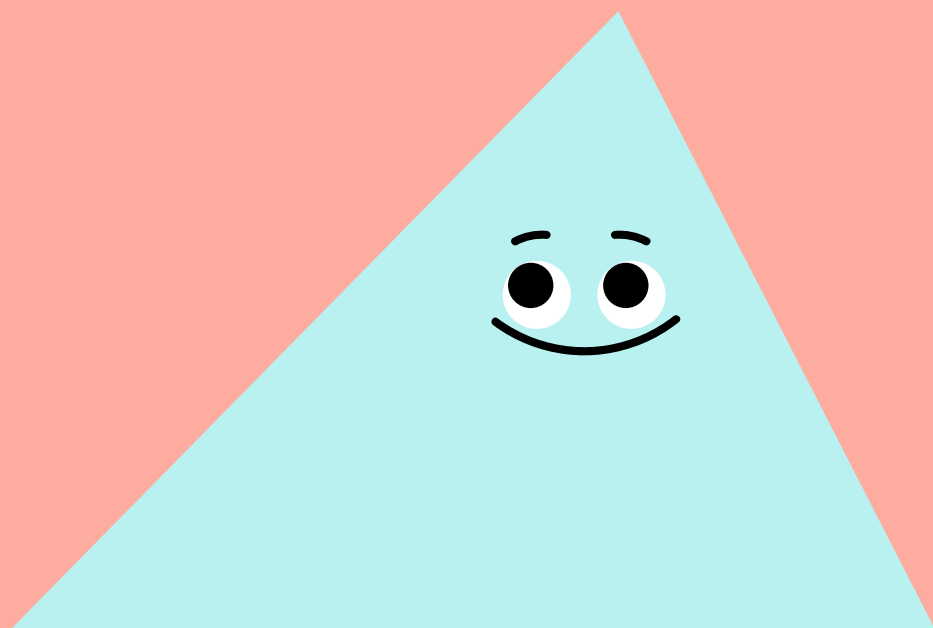
CONCLUSION

- Geometry is everywhere!
 - It helps us understand and create the world around us.
 - Whether in nature, technology, or design, geometry plays a crucial role in shaping our lives.
- 
- 
- 



PROPERTIES OF SHAPES

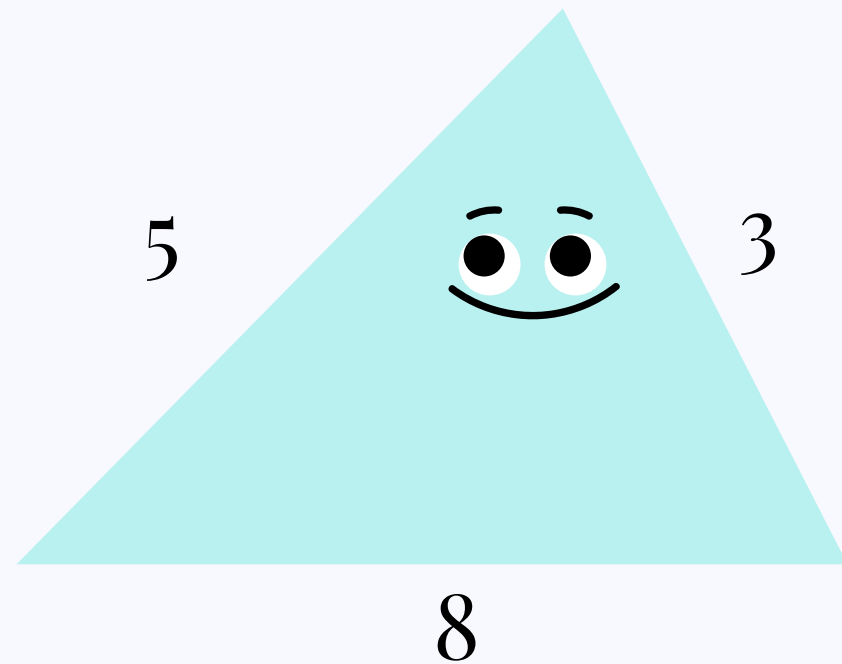
5th Grade Math Presentation



CLASSIFYING TRIANGLES

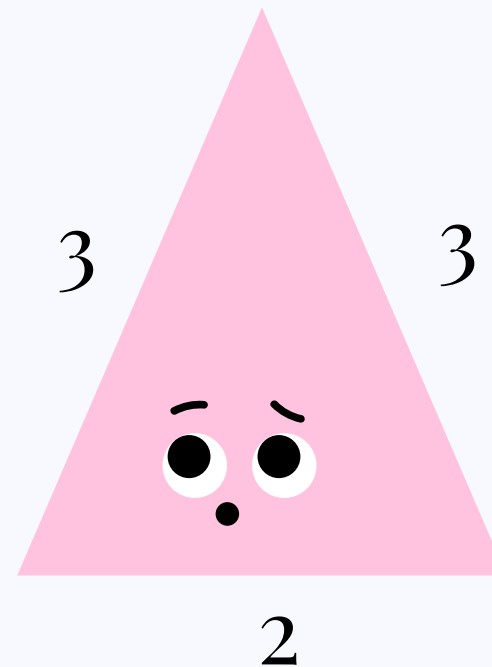
Measure of Sides

SCALENE



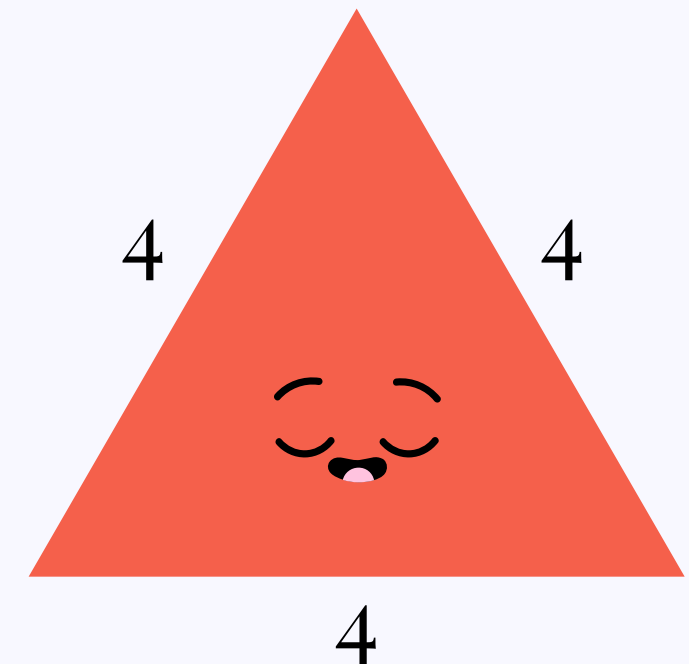
no sides equal

ISOSCELES



at least two sides equal

EQUILATERAL

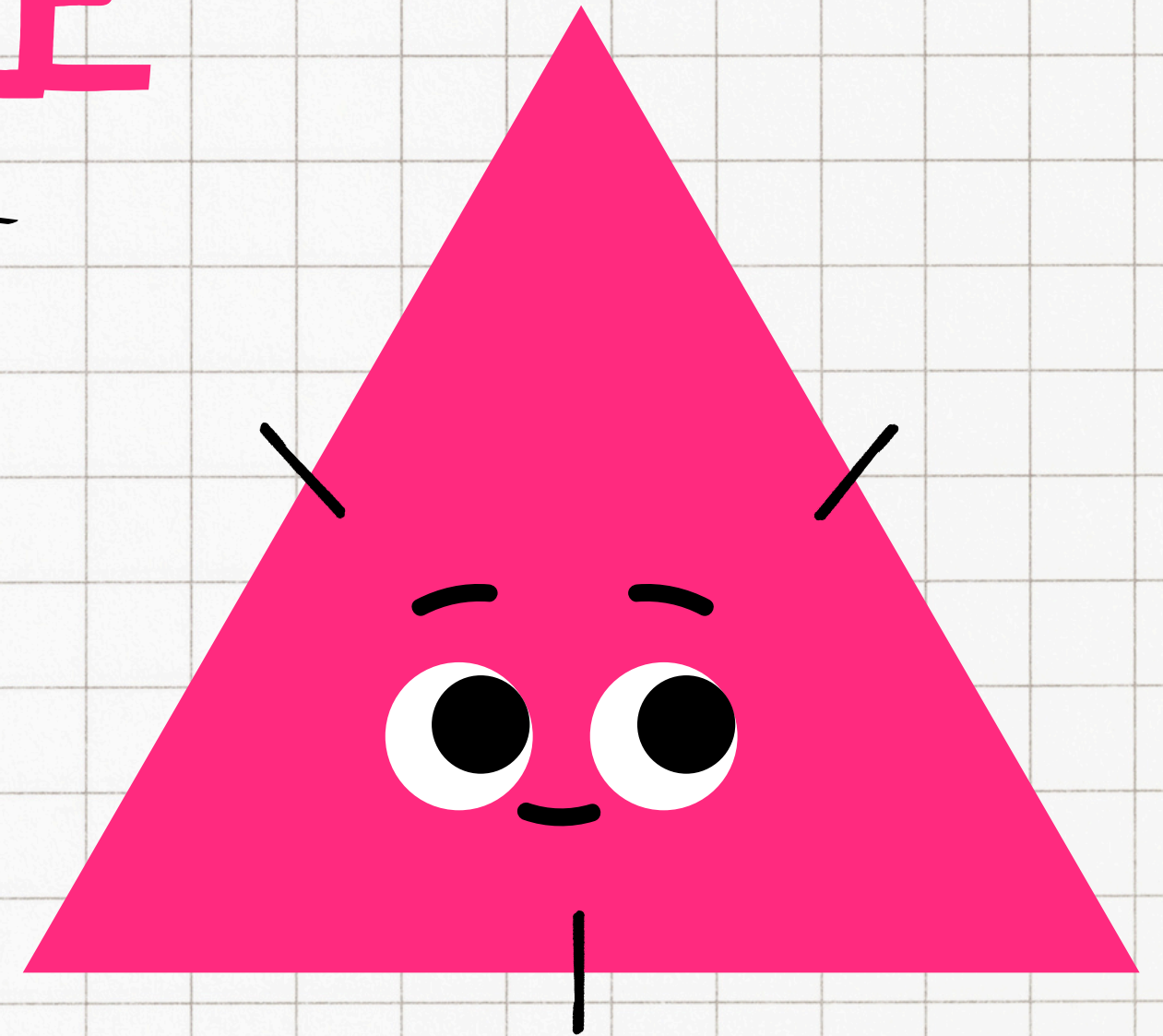


all sides equal

EQUILATERAL TRIANGLE

Definition: All three sides are of equal length.

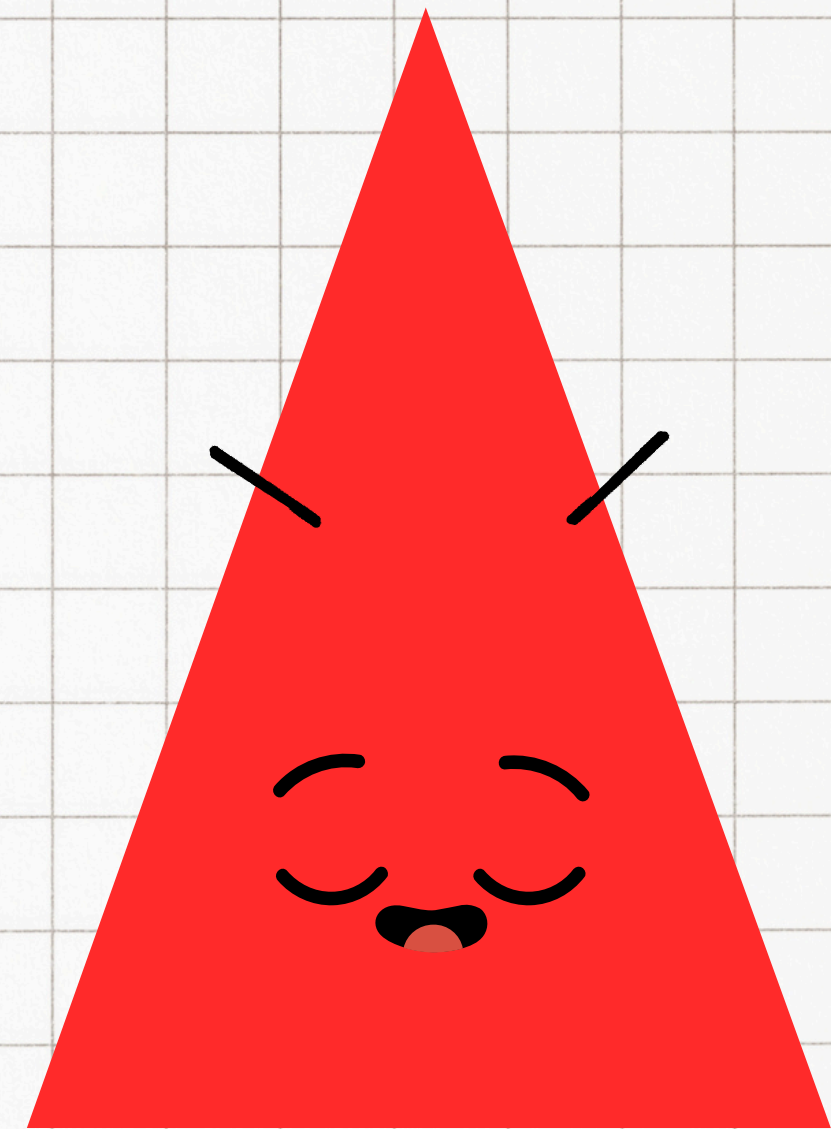
Properties: All three angles are equal, and each angle measures 60° .



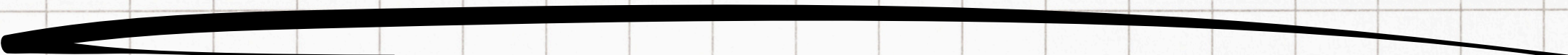
ISOSCELES TRIANGLE

Definition: Two sides are of equal length.

Properties: The angles opposite the equal sides are also equal.

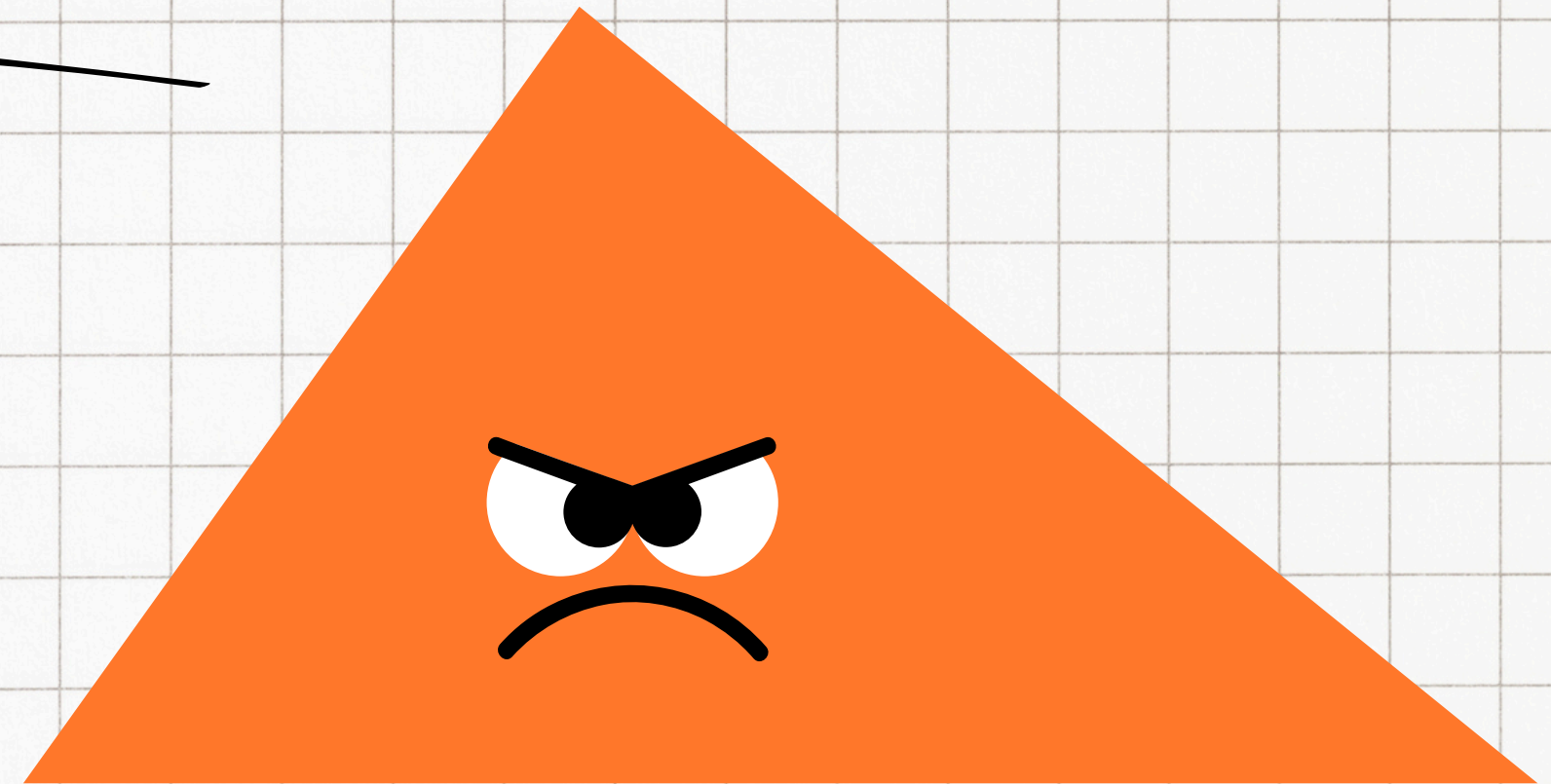


SCALED TRIANGLE



Definition: All three sides are of different lengths.

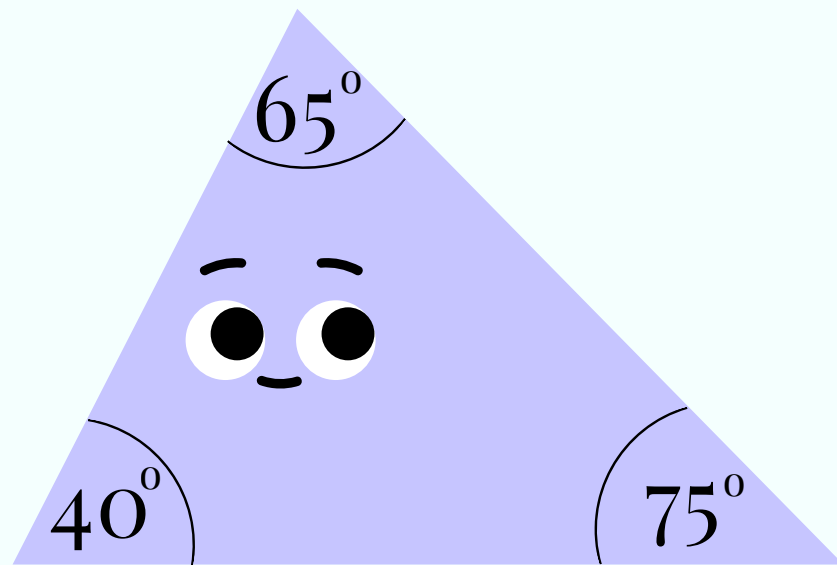
Properties: All three angles are different.



CLASSIFYING TRIANGLES

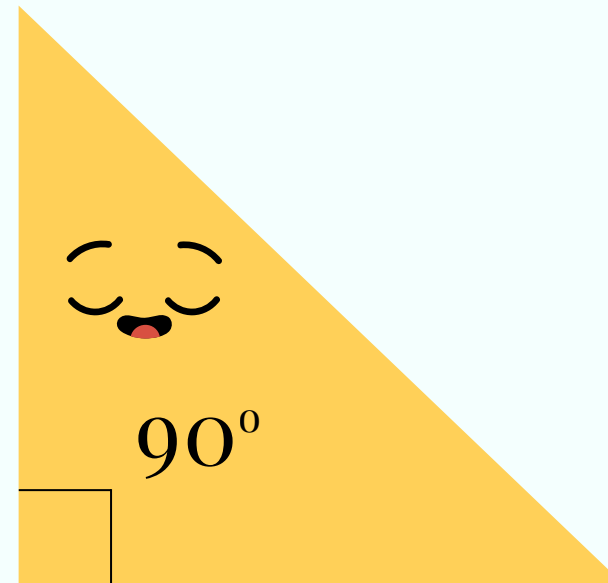
Measure of Angles:

ACUTE



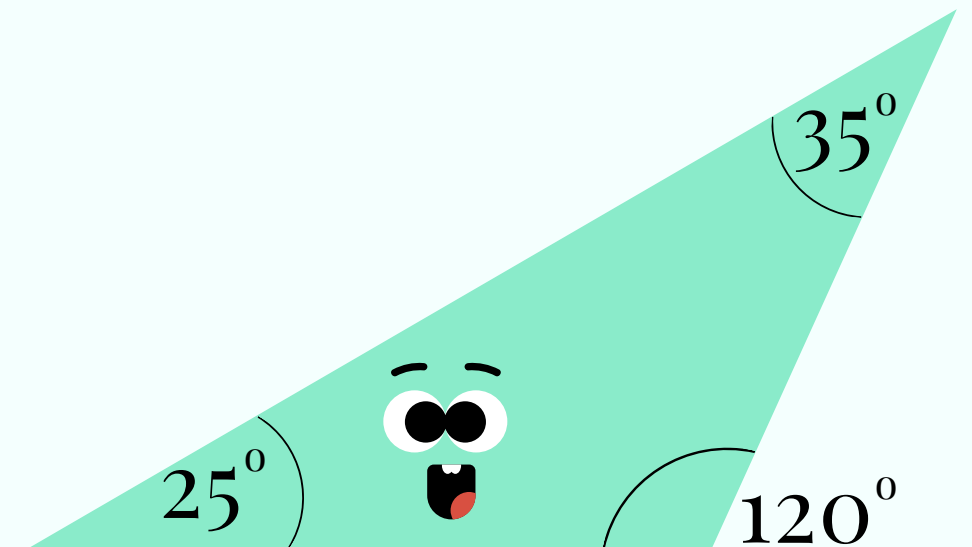
all angles less than 90°

RIGHT



has one angle that is 90°

OBTUSE

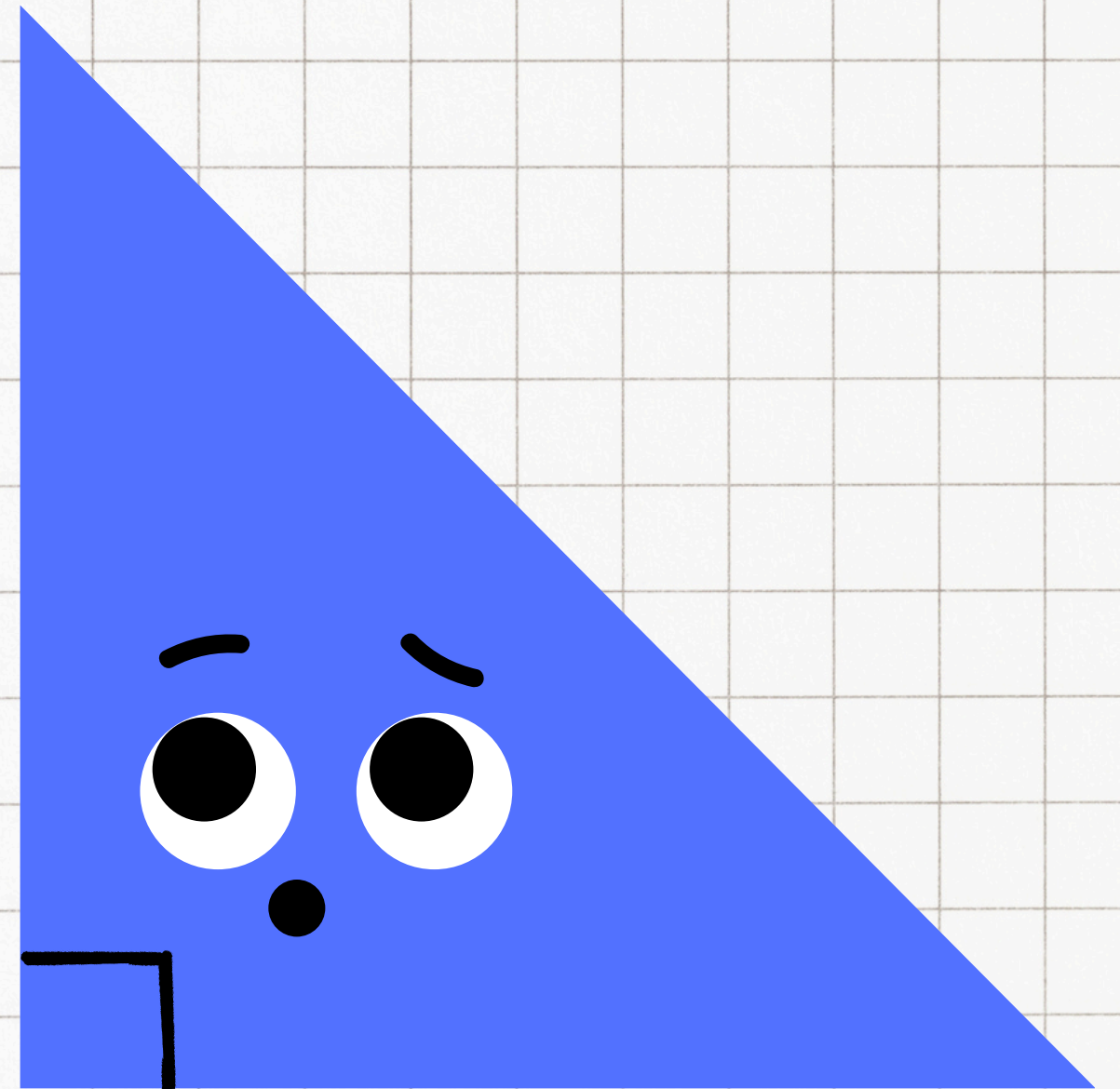


has one angle greater than 90°

RIGHT TRIANGLE

Definition: One of the angles is exactly 90° .

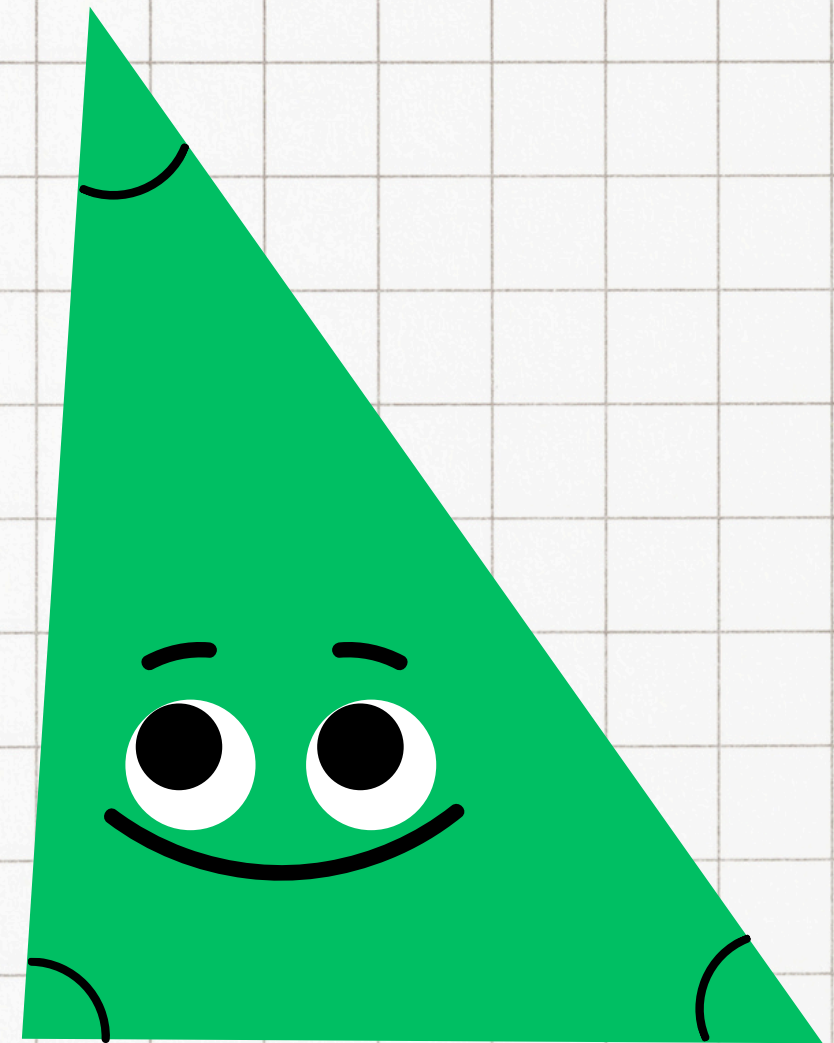
Properties: The side opposite the right angle is the hypotenuse, the longest side of the triangle.



ACUTE TRIANGLE

Definition: All three angles are less than 90° .

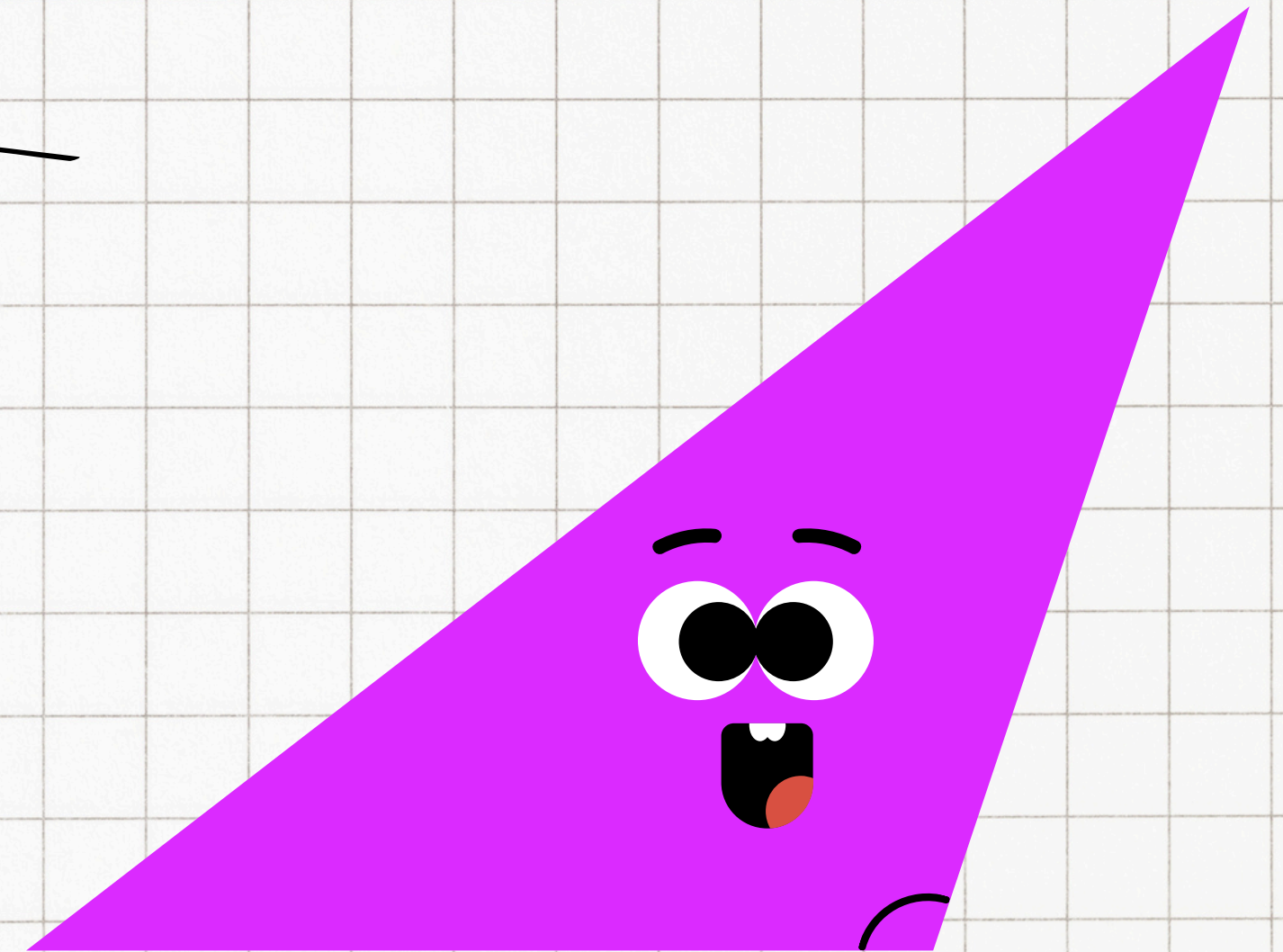
Properties: The triangle is sharp-angled, and the sum of the angles is still 180° .



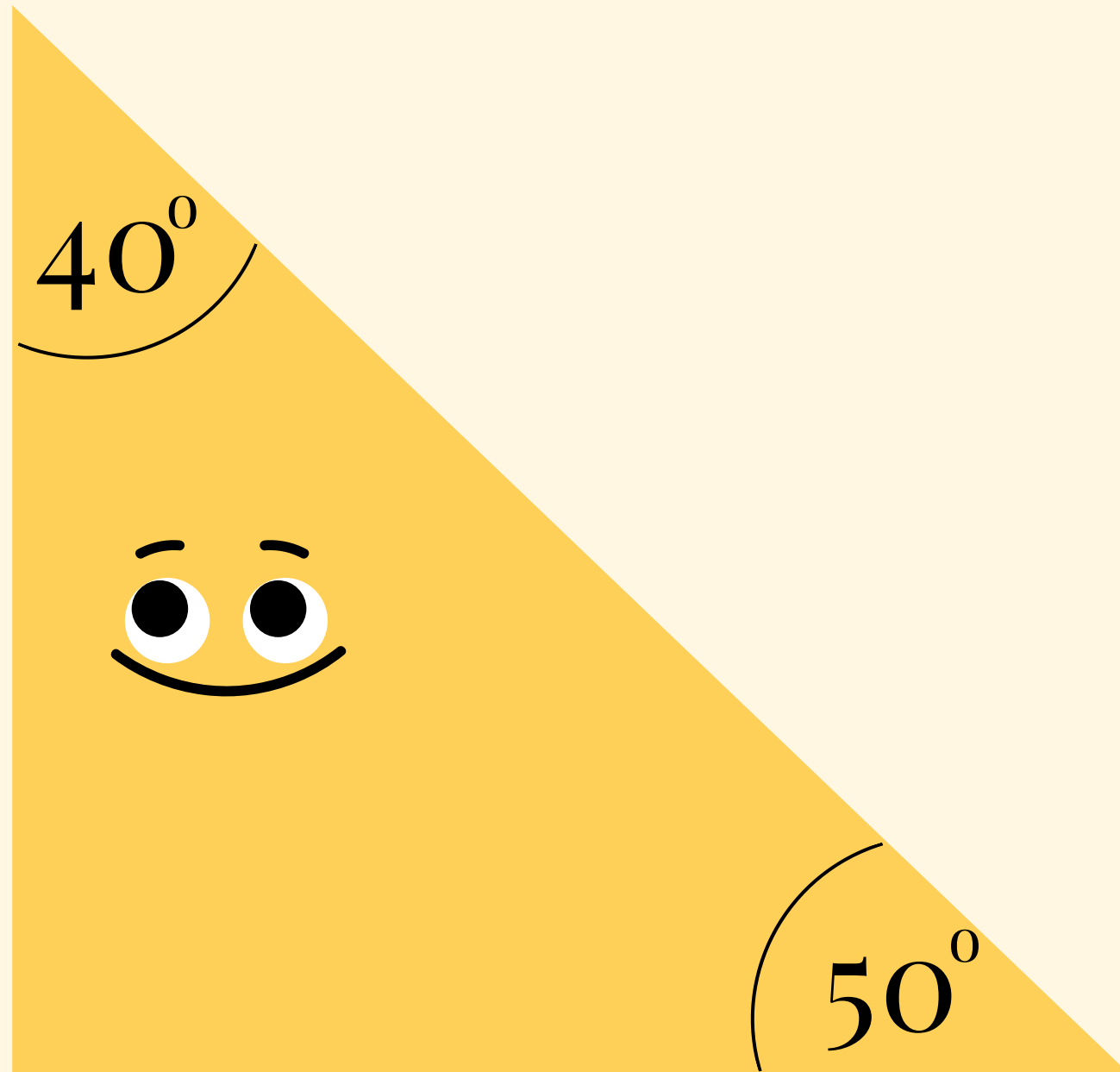
OBTUSE TRIANGLE

Definition: One of the angles is greater than 90° .

Properties: The other two angles are acute (less than 90°).



DEFINING TRIANGLES



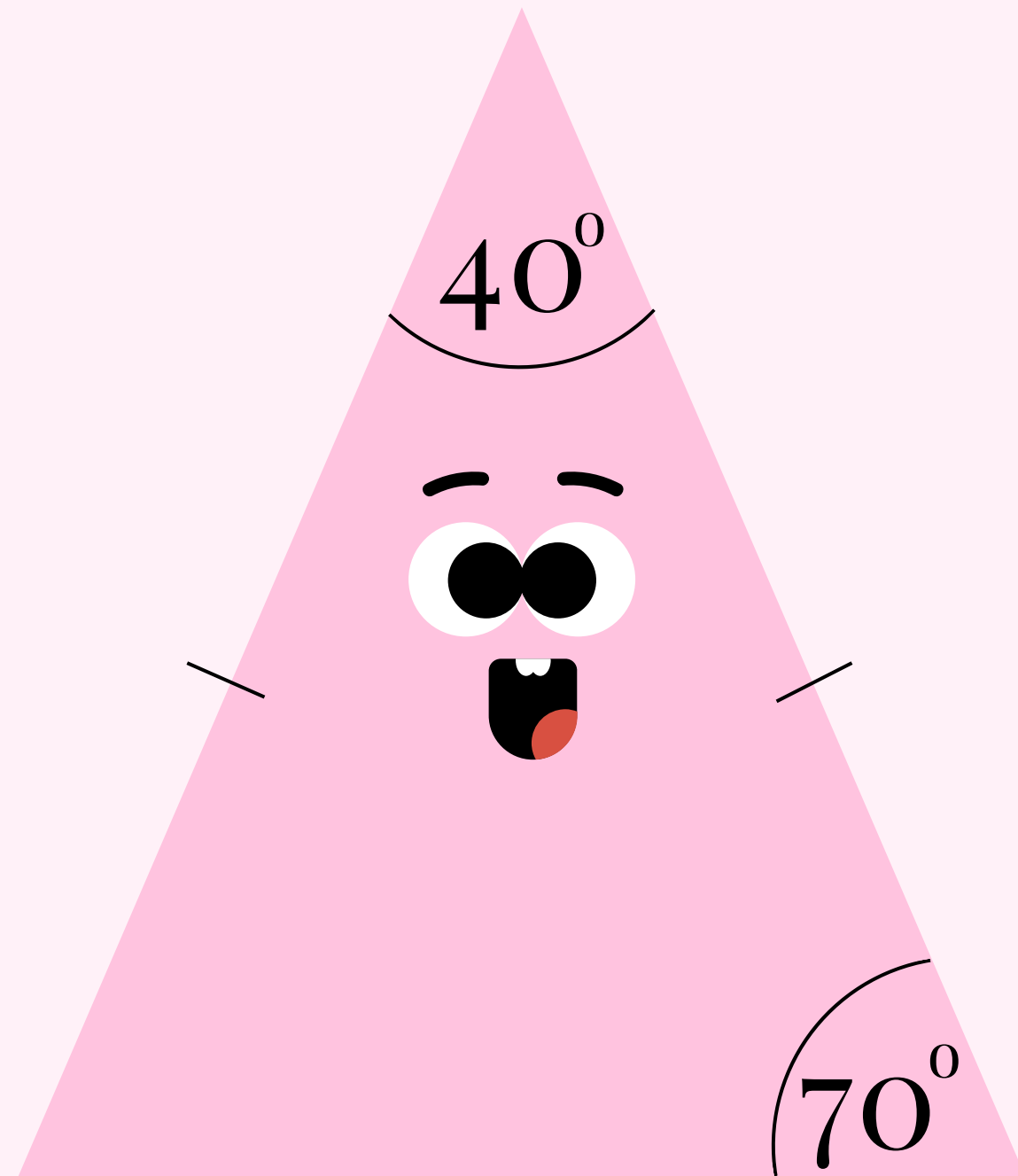
If you know two angles of a triangle, you can always figure out the third. The three angles need to add up to 180 degrees.

$$40 + 50 = 90$$

$$180 - 90 = 90$$

Therefore, this is a right triangle and a scalene triangle.

DEFINING TRIANGLES



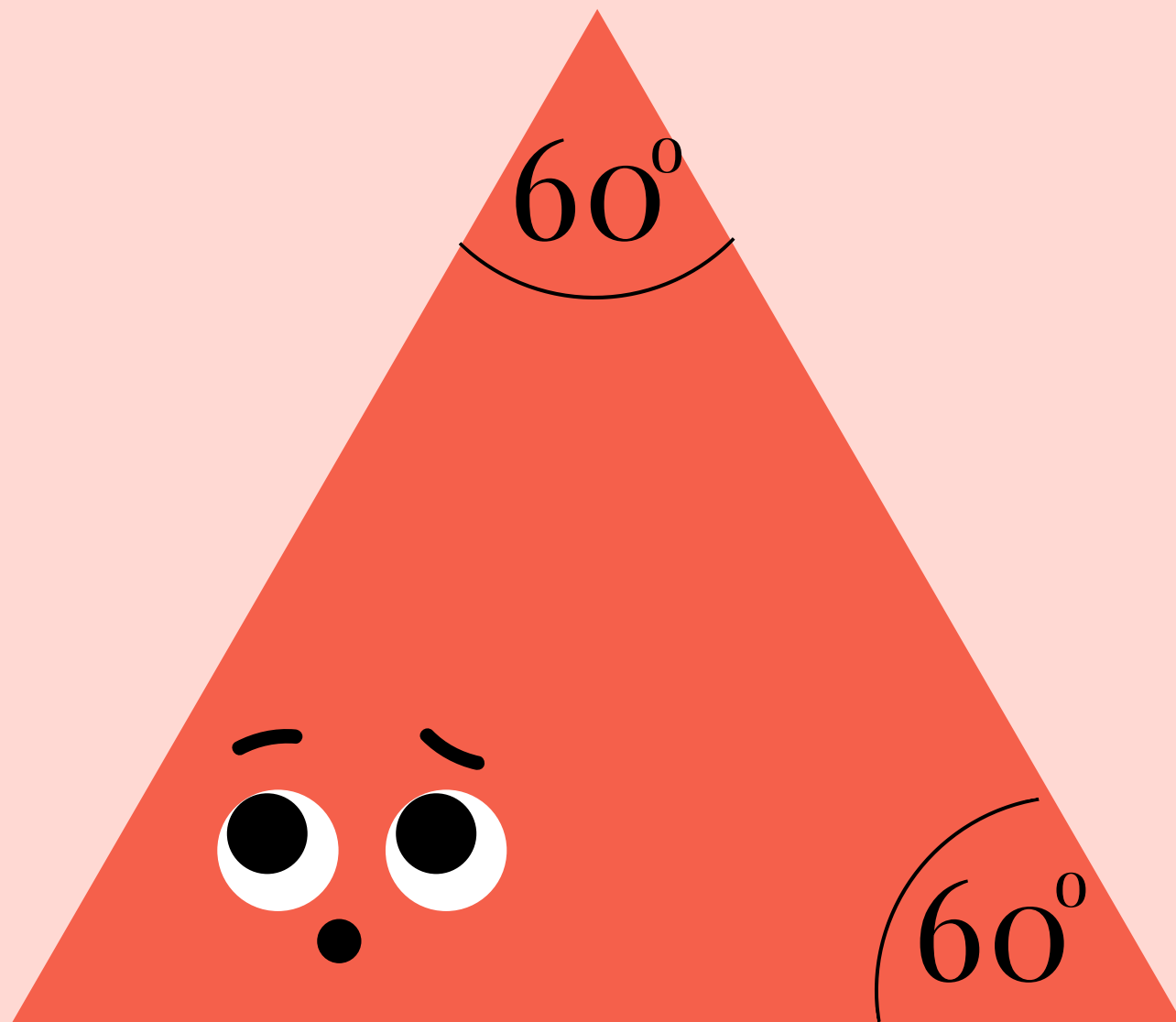
$$40 + 70 = 110$$

$$180 - 110 = 70$$

Therefore, two angles have the same measure. Because their angles are congruent, the opposite sides lengths are also the same.

This is an isosceles triangle.

DEFINING TRIANGLES



$$60 + 60 = 120$$

$$180 - 120 = 60$$

Because all of the angles are congruent, all of the sides are also congruent.

This is an equilateral triangle.

QUADRILATERALS

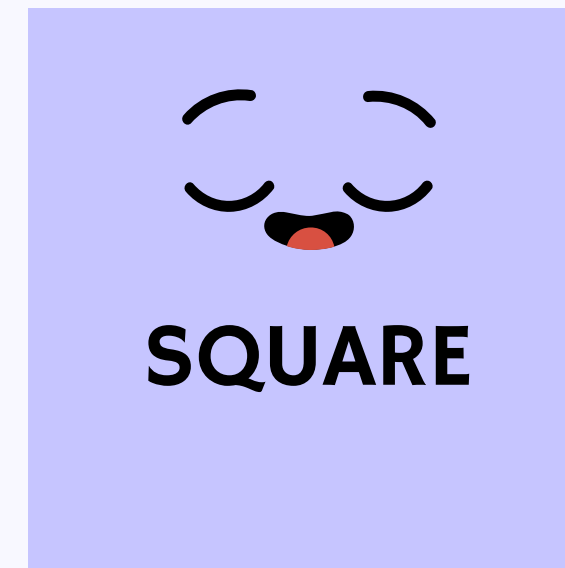
A polygon with four sides, angles, and vertices.



KITE



PARALLELOGRAM



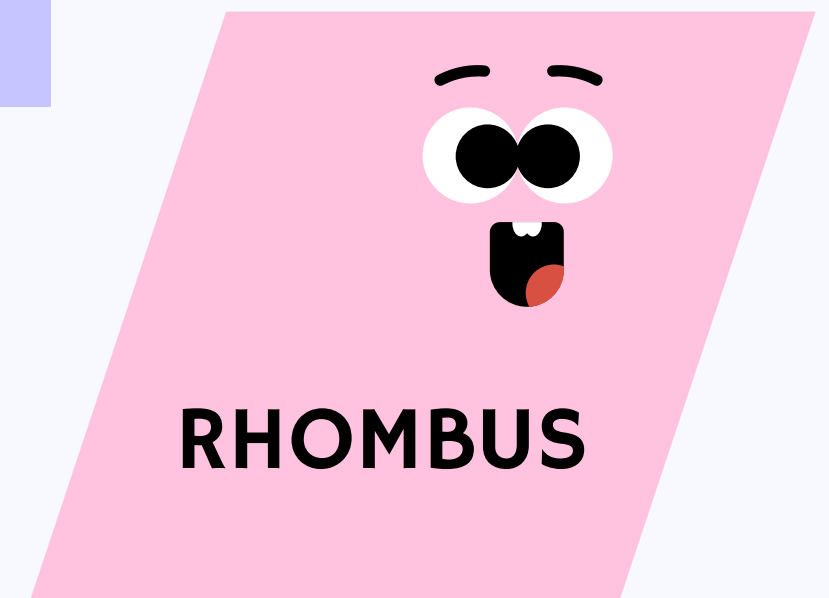
SQUARE



TRAPEZOID



RECTANGLE



RHOMBUS

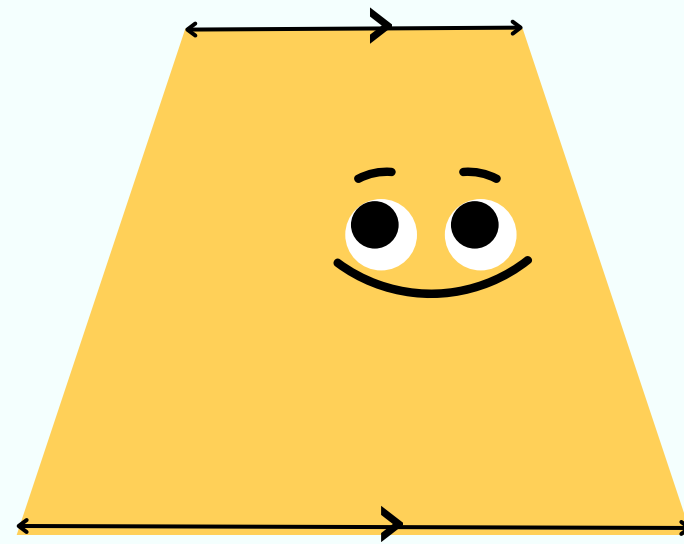
CLASSIFYING QUADRILATERALS

KITES



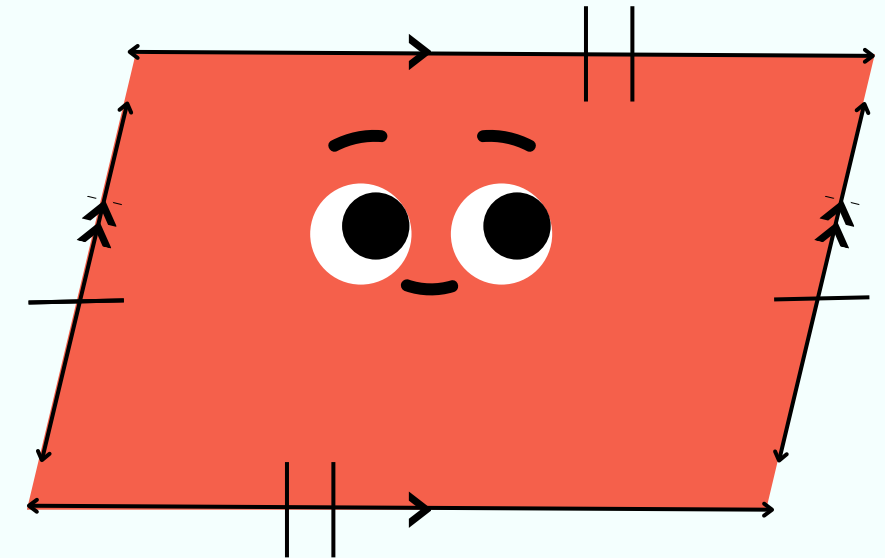
no parallel sides

TRAPEZOIDS



1 pair of parallel sides

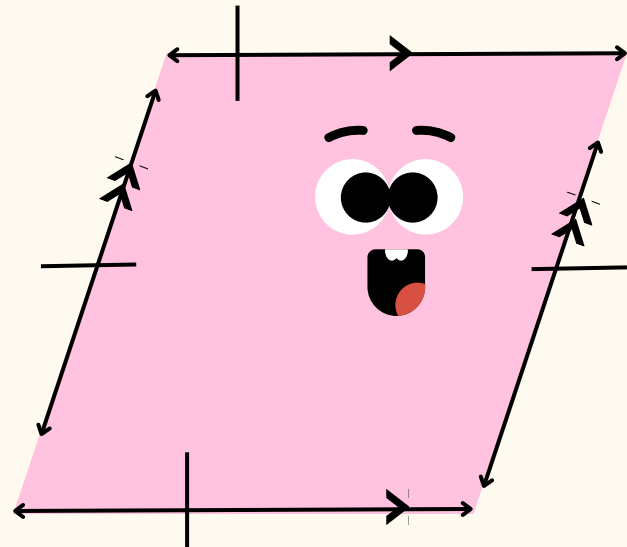
PARALLELOGRAMS



2 pairs of parallel sides

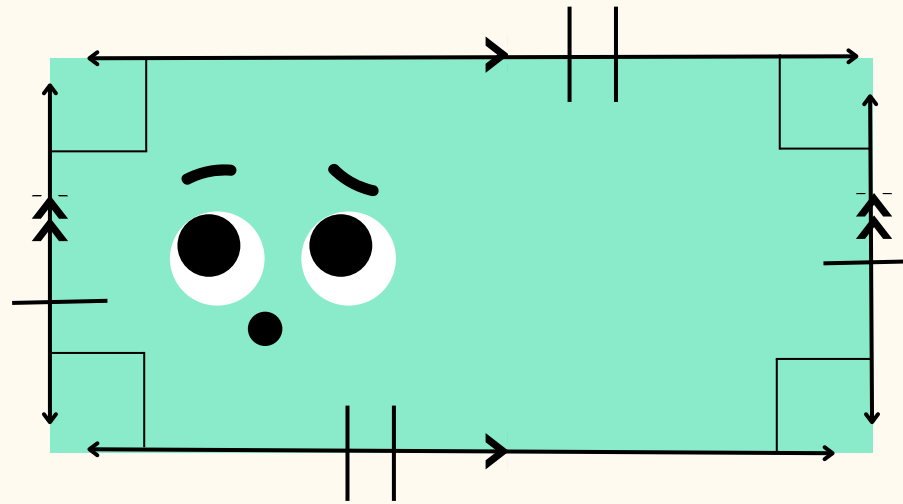
CLASSIFYING QUADRILATERALS

RHOMBUS



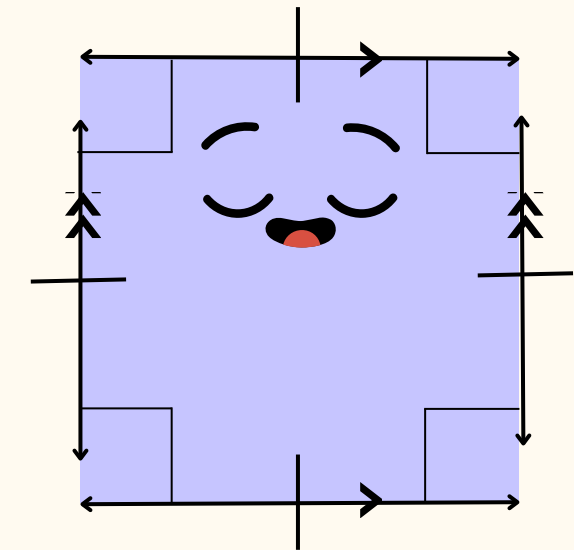
2 pairs of parallel sides
equal sides

RECTANGLES



2 pairs of parallel sides
4 right angles

SQUARES



2 pairs of parallel sides
4 right angles
equal sides

QUADRILATERALS

```
graph TD; A[QUADRILATERALS] --> B[KITE]; A --> C[PARALLELOGRAM]; A --> D[TRAPEZOID]; C --> E[RHOMBUS]; C --> F[RECTANGLE]; E --> G[SQUARE]; F --> G
```

KITE

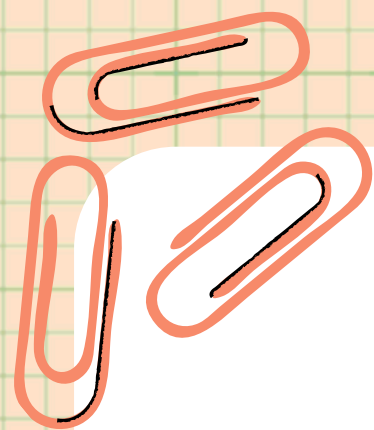
PARALLELOGRAM

TRAPEZOID

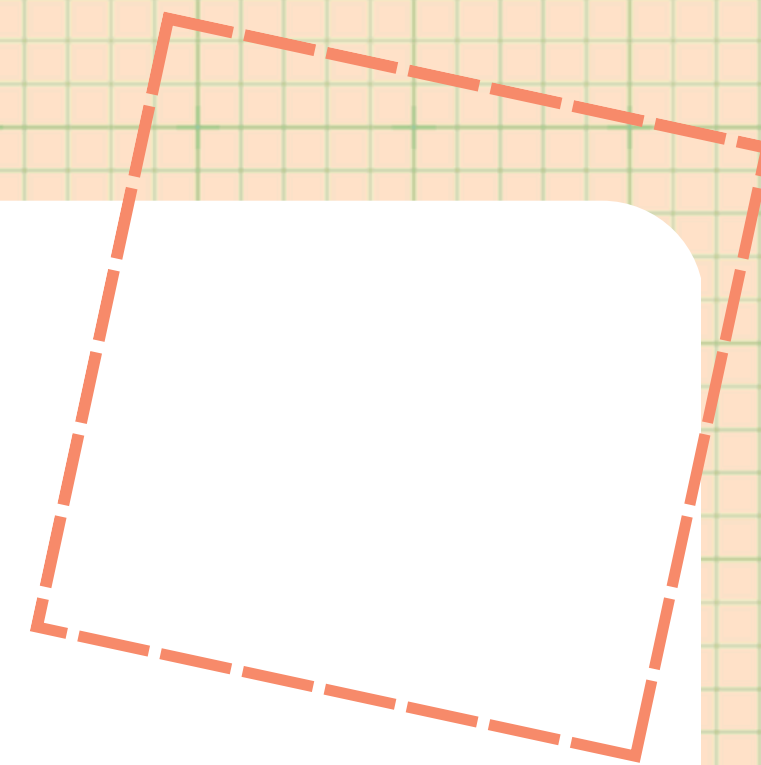
RHOMBUS

RECTANGLE

SQUARE

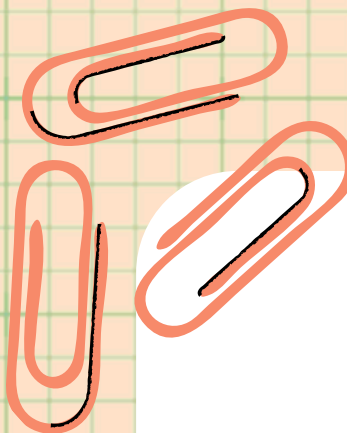


Polygon



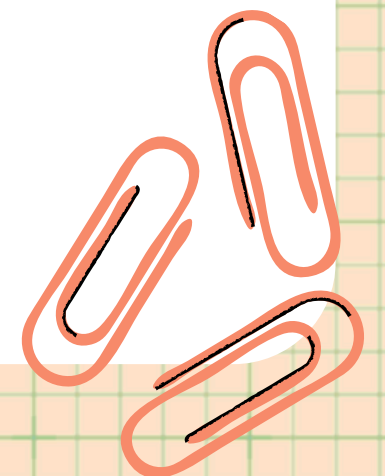
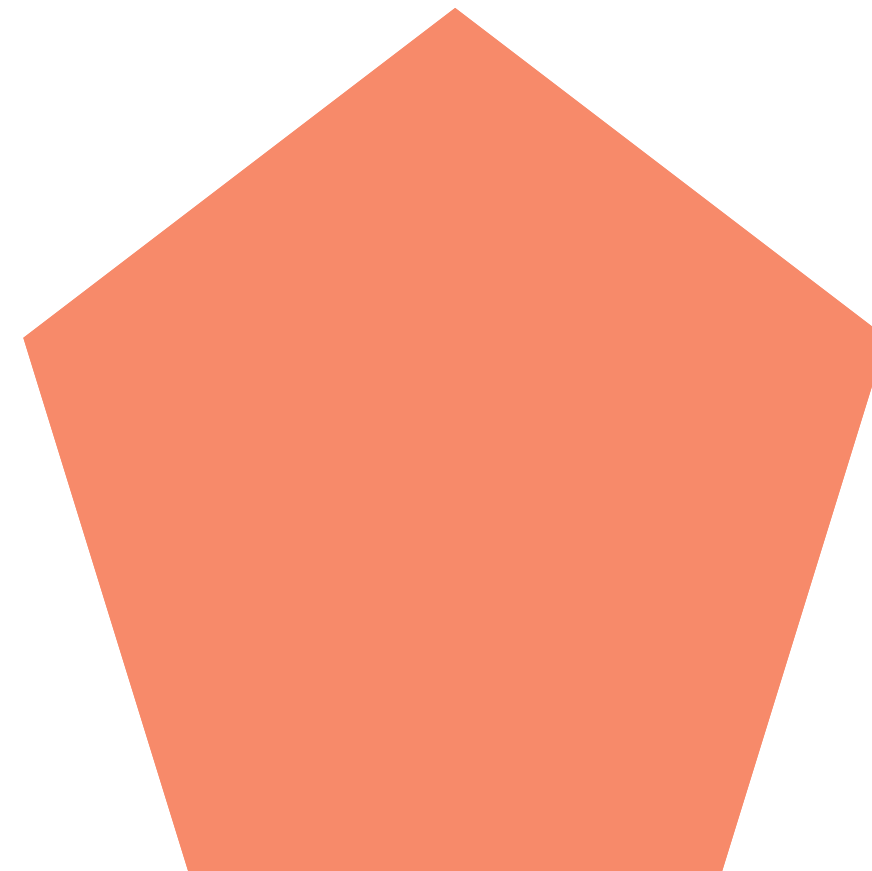
A polygon is a closed figure with straight sides.

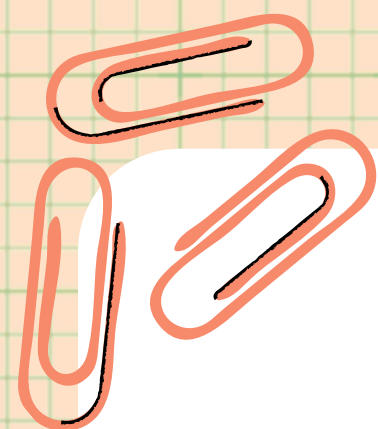
A regular polygon has all equal side lengths and all equal angle measures.



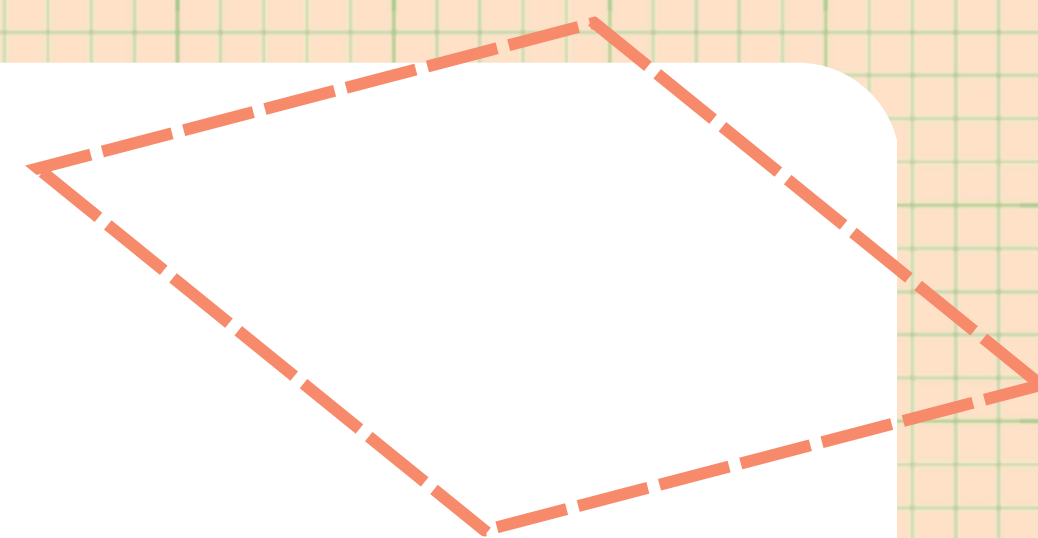
Regular Polygon

This pentagon is a regular polygon because it has equal sides and equal angles.

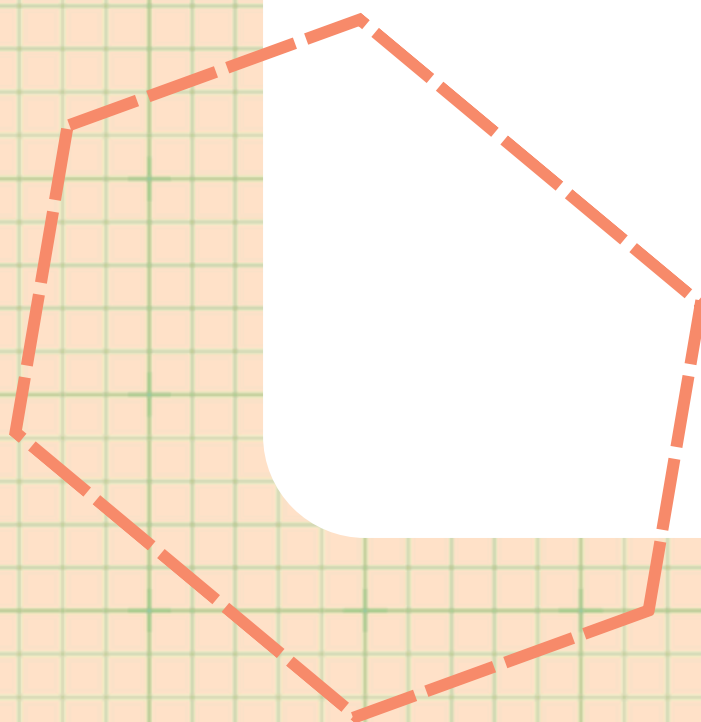


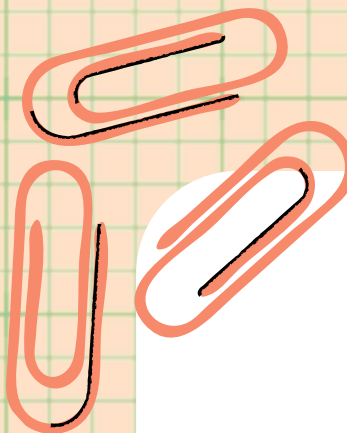


Irregular



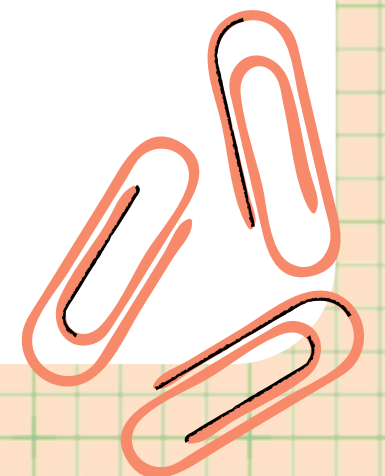
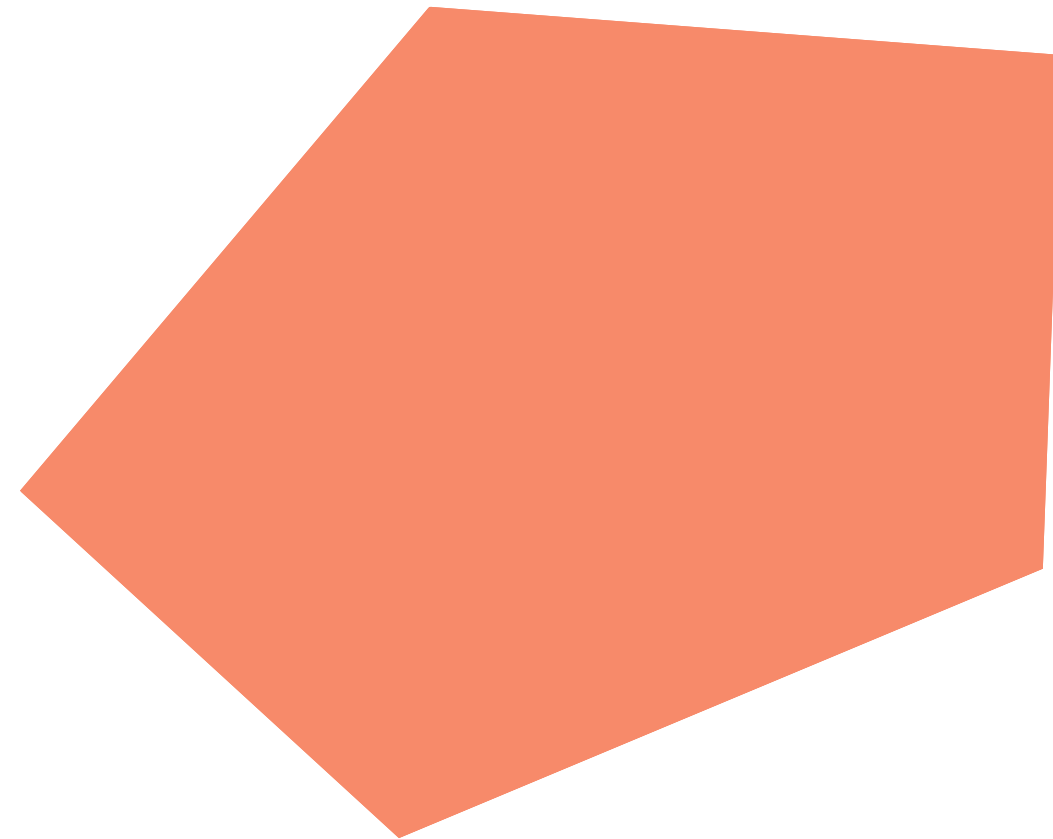
If the sides and angles of a polygon
are not equal (not the same), the
polygon is irregular.

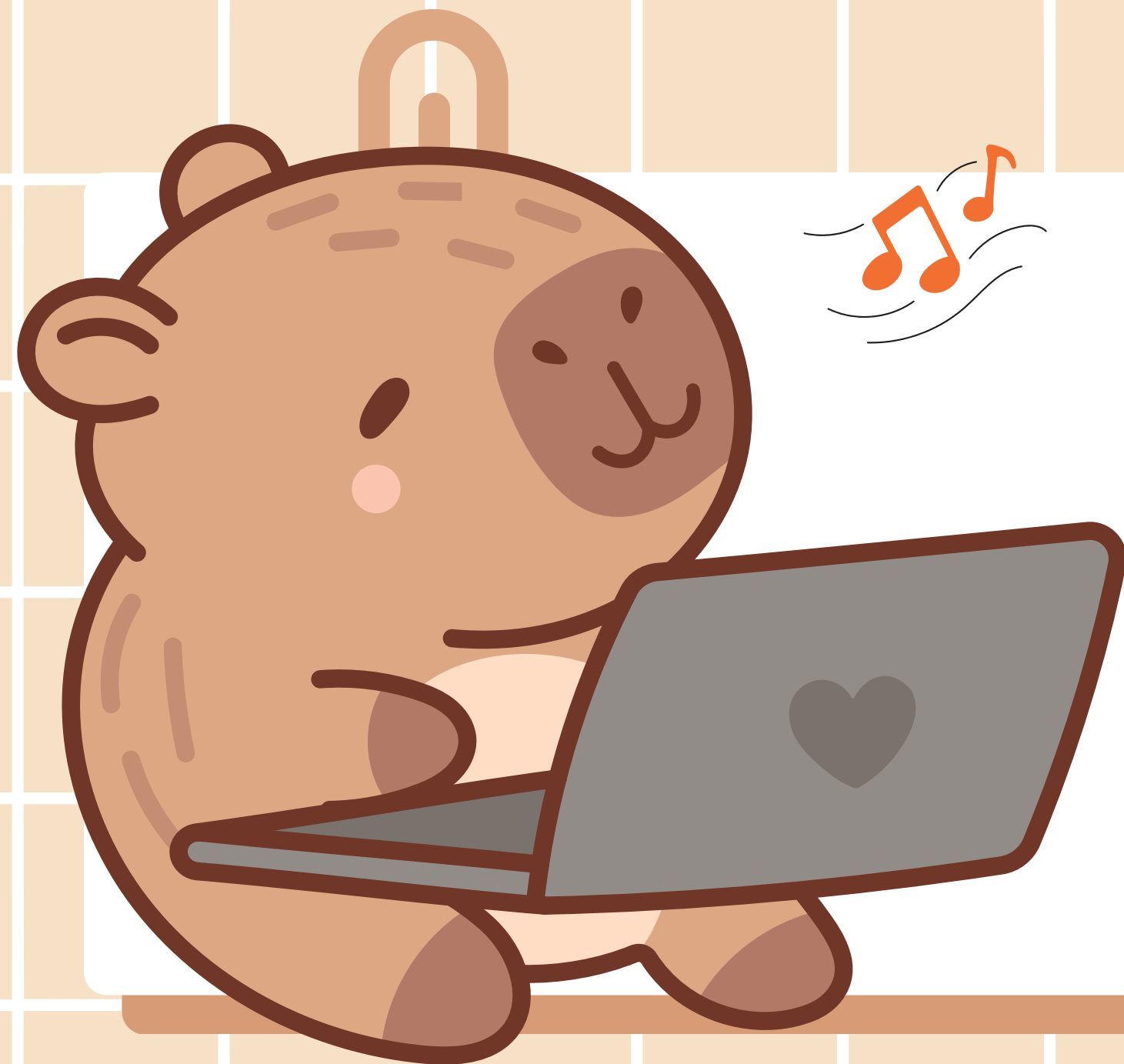




Irregular Polygon

This pentagon is an irregular polygon because it does not have equal sides and equal angles.





Thank
You